Chapter-13

ELASTICITY

- Elasticity is the property of material by virtue of which materials regains its original state (i.e., shape & size) when deforming force is removed.
  - On increasing **temperature** elasticity **decreases.**
  - On **rolling** and **hammering** elasticity **increases.**
  - On **annealing** elasticity decreases.
  - Due to impurity elasticity of material changes.
  - If impurity is more elastic then material then elasticity increases and vice-versa.
  - **Quartz** is consider to be perfectly elastic materials.
  - **Putty** is consider to be perfectly plastic material.

Hooke's law: within elastic limit,

$$\text{Stress} \propto \text{stain} \Rightarrow E = \frac{\text{Stress}}{\text{Strain}}$$

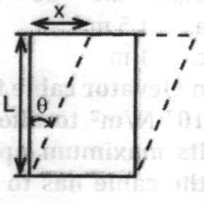
where E = modulus of elasticity,

$$\text{Stress} = \frac{\text{Force}}{\text{Area}} = \frac{F}{A}\ \text{N/m}^2, [\text{Stress}] = ML^{-1}T^{-2}$$

Strain:

(i) Longitudinal strain,

$$= \frac{\text{change in lengths}}{\text{original length}} = \frac{\Delta L}{L}$$



(ii) Volumetric strain,

$$= \frac{\text{change in volume}}{\text{original volume}} = \frac{\Delta V}{V}$$

(iii) Shear strain, $(\theta) = \frac{x}{L}$

(iv) Thermal strain:

It is the longitudinal strain due to change in temperature,

Thermal strain = $\alpha\,\Delta T$

where α = linear expansion coefficient.

ΔT = change in temperature

- **Thermal stress:** If the ends of rod are rigidly fixed and its temperature is changed, then stress developed in the rod are thermal stress.

  Thermal stress = $Y\,\alpha\,\Delta T$

  Y = Youngs modulus elasticity
- Young modulus of elasticity:

$$Y = \frac{\text{Longitudinal stress}}{\text{Longitudinal strain}} = \frac{\frac{F}{A}}{\left(\frac{\Delta L}{L}\right)}$$

(1) For loaded wire,

$$\Delta L = \frac{FL}{\pi r^2 Y}$$

- For right body, $\Delta L = 0 \Leftrightarrow Y = \infty$ elasticity of rigid body is infinite.

(2) If same stretching force is applied to different wire of same material:

$$\Delta L \propto \frac{L}{r^2}\ [\text{F and Y are constant}]$$

- Elongation of wire of length 'L' having material of density 'ρ' due to its **own weight**

$$\Delta L = \frac{L^2 \rho g}{2Y}$$

- Maximum length of elastic wire of density 'ρ' which can support its own weight if breaking strength of the wire is 'f' is,

$$L_{max} = \frac{f}{\rho g}$$

- Work done or potential energy stored in a wire.

$$W = \frac{1}{2} \times \text{force} \times \text{elongation}$$

- Energy density $= \frac{1}{2} \times \text{stress} \times \text{strain}$

$$= \frac{\text{stress}^2}{2Y}$$

$$= \frac{1}{2} \times Y \times \text{strain}^2$$

Bulk modulus of elasticity (k):

$$K = \frac{\text{Normal stress}}{\text{Volumetre strain}} = \frac{\left(\frac{F}{A}\right)}{\left(\frac{\Delta V}{V}\right)} = \frac{\Delta P}{\left(\frac{\Delta V}{V}\right)}$$

- $K = \frac{100 \times \text{change in pressure}}{\%\ \text{decrease in volume}}$
- A material is taken to depth 'h' in a liquid so that it volume decrease by x% then,

$$h = \frac{k\,x}{100\,\rho \times g}\quad \text{For water, } h = kx\ 10^{-6}\text{m}$$

- Compressibility $= \frac{1}{k}$

Isothermal elasticity, $k = \rho$

Adiabatic elasticity, $k = \gamma\rho$

γ = adiabatic exponent,

Rigidity modulus of elasticity (η):

$$\eta = \frac{\text{Tangential stress}}{\text{Stress strain}} = \frac{\left(\frac{F}{A}\right)}{\theta}$$

- For liquid and gas modulus of rigidity is zero.
- Poission's Ratio (σ):

$$\sigma = \frac{\text{Lateral strain}}{\text{Longitudinal strain}}$$

$$\sigma = -\frac{\left(\frac{\Delta d}{d}\right)}{\left(\frac{\Delta L}{L}\right)} = -\frac{\left(\frac{\Delta r}{r}\right)}{\frac{\Delta L}{L}}$$

- Theoretical value of σ lies between –1 and 0.5
- Practical value of σ lies between, 0 and 0.5
- Relation between volumetric stain and longitudinal strain,

$$\frac{\Delta V}{V} = (1 - 2\sigma)\frac{\Delta L}{L}$$

- If $\sigma = 0.5$, $\Delta V = 0$ there is are change in volume

Relation among γ, κ and η

- $Y = 3\kappa(1 - 2\sigma)$
- $Y = 2\eta(1 + \sigma)$
- $\frac{9}{Y} = \frac{1}{\kappa} + \frac{3}{\eta}$
- Breaking stress is constant for a material.
- Breaking stress does not depend on length of material.

  Breaking force $\propto A \propto r^2$
- Work done in elongation of spring from ℓ_1 to ℓ_2

  $W = \frac{1}{2}\kappa(\ell_2^2 - \ell_1^2)$

  Force constant, $\kappa = Y r_0$ where, r_0 = equilibrium inter molecular distance
- $F = -\frac{du}{dr}$, [u = potential energy]

(i) At intermolecular separation $r = r_0$, PE is minimum

i.e., F = 0

(ii) If $r > r_0$; Force is attractive, PE = –ve

(iii) If $r < r_0$; Force is repulsive, PE = +ve

Multiple Choice Questions

1. A uniform elastic plank moves due to a constant force F distributed uniformly over the end face. The area of end face is S and Young's modulus of the material is E. What is average strain produced in the direction of the force?

a. $\frac{F}{SE}$ b. $\frac{F}{2SE}$

c. $\frac{F}{4SE}$ d. zero

2. What is the change in density of water in a lake at a depth of 400 m below the surface ? The density of water at the surface is 1030 kg/m³ and bulk modulus of water is 2×10^9 N/m² :

a. 4 kg/m³ b. 2 kg/m³

c. 10.2 kg/m³ d. 102.6 kg/m³

3. A wire is elongated by 2 mm when a brick is suspended from it. When the brick is immersed in water, the wire contracts by 0.6 mm. What is the density of brick?

(Density of water = 1000 kg/m³)

a. 3333 kg/m³ b. 4210 kg/m³

c. 5000 kg/m³ d. 2000 kg/m³

4. A metal block is experiencing an atmospheric pressure of 10^5 N/m². When the same block is placed in a vacuum chamber, the fractional change in its volume (the bulk modulus of metal is 1.25×10^{11} N/m²) :

a. 4×10^{-7} b. 2×10^{-7}

c. 8×10^{-7} d. 1×10^{-7}

5. Four wires of the same material are stretched by the same load. The dimensions of the wives are as given below. The one which has the maximum elongation is of :

a. diameter 1 mm and length 1 m

b. diameter 2 mm and length 2 m

c. diameter 0.5 m and length 0.2 m

d. diameter 3 mm and length 3 m

6. A brass rod of length 2 m and cross-sectional area 2.0 cm² is attached end to end to a steel rod of length L and cross-sectional area 1.0 cm². The compound rod is subjected to equal and opposite pulls of magnitude 5×10^4 N at its ends. If the elongations of the two rods are equal, the length of the steel rod (L) is ($Y_{Brass} = 1.0 \times 10^{11}$ N/m² and $Y_{steel} = 2.0 \times 10^{11}$ N/m²)

a. 1.5 m b. 1.8 m

c. 1 m d. 2 m

7. An elevator cable is to have a maximum stress of 7×10^7 N/m² to allow for appropriate safety factors. Its maximum upward acceleration is 1.5 m/s². If the cable has to support the total weight of 2000 kg of a loaded elevator the area of cross-section of the cable should be:

a. 3.28 cm² b. 2.38 cm²

c. 0.328 cm² d. 8.23 cm²

8. A uniform steel bar of cross-sectional area A and length L is suspended to that it hangs vertically. The stress at the middle point of the bar is (ρ is the density of steel)

a. $\frac{L}{2A}\rho g$ b. $\frac{L\rho g}{2}$

c. $\frac{LA}{\rho g}$ d. $L\rho g$

9. A cable that can support a load W is cut into two equal parts. The maximum load that can be supported by either part is:

a. W/4 b. W/2

c. W d. 2W

10. The Young's modulus of a perfectly rigid body is :

a. is zero b. is unity

c. is infinity

d. may have any finite non-zero value

11. In practice, Poisson's ratio σ lies between:

a. $-\infty$ to $+\infty$ b. 0 and $+\infty$

c. 0 and 0.5 d. 0.5 and 0

12. The length of a metal wire is l_1, when the tension in it is T_1 and is l_2 when the tension is T_2. The unstretched length of the wire is:

a. $\sqrt{l_1 l_2}$ b. $\frac{l_1 + l_2}{2}$

c. $\frac{l_1T_2 - l_2T_1}{T_2 - T_1}$ d. $\frac{l_1T_2 + l_2T_1}{T_2 + T_1}$

13. The Young's modulus of brass and steel are respectively 10×10^{10} N/m² and 2×10^{10} N/m². A brass wire and a steel wire of the same length are extended by 1 mm under the same force; the radii of brass and steel wires are R_B and R_S respectively. Then :

a. $R_S = \sqrt{2}\, R_B$ b. $R_S = \frac{R_B}{\sqrt{2}}$

c. $R_S = 4R_B$ d. $R_S = \frac{R_B}{4}$

14. The length of an elastic string is a metre when the tension is 4 newton, and b metre when the tension is 5 newton. The length in metres when the tension is 9 newton is :

a. 4a – 5b b. 5b – 4a

c. 9b – 9a d. a + b

15. A stress of 10^6 N/m² is required for breaking a material. If the density of the material is 3×10^3 kg/m³, then what should be the length of the wire made of this material so that it breaks under its own weight :

a. 10 m b. 33.3 m

c. 5 m d. 64 m

16. A uniform pressure P is exerted on all sides of a solid cube at temperature t°C. By what amount should the temperature be raised in order to bring the volume back to that it had been before the pressure was applied.
(Linear expansivity of the material of the cube is α and the bulk modulus of elasticity is β)

a. $\frac{P\alpha}{\beta}$ b. $\frac{3P\alpha}{\beta}$

c. $\frac{P}{\alpha\beta}$ d. $\frac{P}{3\alpha\beta}$

17. A wire is suspended by one end. At the other end a weight Mg is attached. The wire extended by l. The ratio of mechanical energy stored in the stretched wire to the work done by Mg is:

a. 1/1 b. 1/3

c. 1/2 d. 2/1

18. Vessel of 1×10^{-3} m³ volume contains an oil. If a pressure of 1.2×10^5 N/m² is applied on it then volume decreases by 0.3×10^{-3} m³. The bulk modulus of oil is :

a. 6×10^{10} N/m² b. 4×10^5 N/m²

c. 2×10^7 N/m² d. 1×10^6 N/m²

19. Which one of the following affects the elasticity of a substance?

a. change in temperature

b. hammering and annealing

c. impurity in substance

d. all of the above

20. A stretched rubber has :

a. decreased potential energy

b. decreased kinetic energy

c. increased kinetic energy

d. increased potential energy

21. A ball falling in a lake of depth 200 m creates a decreases 0.1% in its volume at the bottom. The bulk modulus of the material of that ball will be:

a. 19.6×10^{-8} N/m² b. 19.6×10^{10} N/m²

c. 19.6×10^{-10} N/m² d. 19.6×10^{8} N/m²

22. A bob of mass 10 kg is attached to a wire 0.3 m long. Its breaking stress is 4.8×10^7 N/m². The area of cross-section of the wire is 10^{-6} m². What is the maximum angular velocity with which it can be rotated in a horizontal circle:

a. 8 rad/sec b. 4 rad/sec

c. 2 rad/sec d. 1 rad/sec

23. For a given material the Young's modulus is 2.4 times that of rigidity modulus, the Poisson's ratio is :

a. 0.2 b. 0.4

c. 1.2 d. 2.4

24. A thick copper rope of density 1.5×10^3 kg/m³ and Young's modulus 5×10^6 N/m², 8 m in length is hung from the ceiling of a room. The increase in its length due to its own weight will be:

a. 9.40×10^{-2}m b. 9.6×10^{-3}m

c. 19.2×10^{-7}m d. 9.6×10^{-7} m

25. Two wires of same material and length but diameter in the rate 1:3 are stretched by the same force. The potential energy of unit volume for the two wires when stretched will be in the ratio:

a. 16 : 1 b. 81 : 1

c. 9 : 1 d. none of these

26. A copper rod 2 m long is stretched by 1mm. Y = 1.2×10^{11} N/m² is given. Then the energy stored per unit volume will be:

a. 30×10^3 J/m³ b. 15×10^3 J/m³

c. 7.5×10^3 J/m³ d. none of these

27. The breaking stress of a wire depends upon:

a. material of wire

b. length of wire

c. radius of wire

d. shape of cross-section

28. The Young's modulus of a wire of length (L) and radius (r) is Y. If the length of reduced to $\frac{L}{2}$ and radius $\frac{r}{2}$, then its Young's modulus will be:

a. $\frac{Y}{2}$ b. Y

c. 2Y d. 4Y

Answers:

1. b	2. b	3. a	4. c	5. c
6. d	7. a	8. b	9. c	10. c
11. c	12. c	13. b	14. b	15. b
16. d	17. c	18. b	19. d	20. d
21. d	22. b	23. a	24. a	25. b
26. b	27. a	28. b		

Solution for Selected Problems

2. $K = \frac{\Delta P}{(\Delta\rho/\rho)} \Rightarrow \Delta\rho = \frac{\Delta P \times \rho}{k} = \frac{h\rho g \times \rho}{k}$

$= \frac{400 \times 1030 \times 9.8 \times 1030}{2 \times 10^9}$

$\therefore \Delta\rho = 2.07\ kg/m^3$

$\simeq 2\ kg/m^3$

3. Force (wt.) $\propto$ elongation

when brick is immersed in water then its weight decreases due to upthrust.

$\frac{\text{wt. in Air}}{\text{wt. in water}} = \frac{\text{elongation in Air}}{\text{elongation in water}}$

$\Rightarrow \frac{\rho_b}{\rho_b - \rho_w} = \frac{2}{1.4}$ $\therefore \rho_b = 3333.3 kg/m^3$

4. $K = \frac{\Delta\rho}{-(\Delta v/v)} = \frac{0 - 10^5}{-(\Delta v/V)}$

$\therefore \frac{\Delta V}{V} = 10^5/1.25 \times 10^{11} = 8 \times 10^{-7}$

6. $\Delta L = \frac{F.L}{AY}$

$\therefore \Delta L_1 = \Delta L_2 \Rightarrow \frac{5 \times 10^4 \times L}{2 \times 10^{11} \times 1} = \frac{5 \times 10^4 \times 2}{1 \times 10^{11} \times 2}$

$\Rightarrow L = 2m$

7. Tension in the cube : $T = Mg + Ma = M(g + a)$
$= 2000 \times 11.5 = 23000$

stress $= \frac{T}{A}$ $\therefore$ $A = \frac{23000}{(7 \times 10^7)} = 3.28\ cm^2$

13. $\Delta L = \frac{FL}{AY} = \frac{F.L}{\pi R^2 Y}$

$\Rightarrow \frac{F.L}{\pi R_B^2 \times Y_B} = \frac{F \times L}{\pi \times R_s^2 \times Y_s} \Rightarrow R_B = \sqrt{\frac{Y_s}{Y_B}} \times R_s$

$\Rightarrow R_B = \sqrt{2}\, R_s$ $\therefore R_s = \frac{R_B}{\sqrt{2}}$

14. $F \propto$ elongation (ΔL) $\therefore \frac{F}{\Delta L}$ = constant

$\frac{4}{a - l_o} = \frac{5}{b - l_o} = \frac{9}{c - l_o}$

[where, l_o = unstretched length and c = length when tension is 9N]

on solving we get.

$c = 5b - 4a$

15. $L = \frac{f}{\rho g} = \frac{10^6}{3 \times 10^3 \times 10} = 33.3m$

16. $\gamma = 3\alpha$; $K = \beta$ $\gamma = \frac{\Delta v}{v \times t}$ $\therefore$ $\frac{\Delta v}{v} = \gamma t$

$K = \frac{\Delta P}{\left(\frac{\Delta v}{v}\right)} = \frac{P}{\gamma t} = \frac{P}{3\alpha t} \Rightarrow t = \frac{P}{3\alpha\beta}$

17. Mechanical energy(U) $= \frac{1}{2} \times Mg \times l$

Work done by Mg : $w = Mg \times l$

Ratio of U to W $= \frac{1}{2}$

18. $K = \frac{\Delta P}{-\left(\frac{\Delta v}{v}\right)} = \frac{1.2 \times 10^5}{\left(\frac{0.3 \times 10^{-3}}{1 \times 10^{-3}}\right)} = 4 \times 10^5\ N/m^2$

21. $K = \frac{\Delta P}{-\left(\frac{\Delta v}{v}\right)} = \frac{100 \times \Delta P}{\%\ \text{decrease in volume}} = \frac{100 \times h\rho g}{0.1}$

$\therefore K = 19.6 \times 10^8\ N/m^2$

22. Tension in the wire $= 4.8 \times 10^7 \times 10^{-6} = 48$ N

$T = m\omega^2 r \Rightarrow 48 = 10 \times \omega^2 \times 0.3$

$\omega = 4$ rad

23. $Y = 2x(1 + \sigma) \Rightarrow 2.4x = 2x(1 + \sigma)$

$\therefore \sigma = \left(\frac{2.4}{2}\right) - 1 = 0.2$

24. Increase in wire due to its own weight

$\Delta L = \frac{L^2\rho g}{2Y} = \frac{8^2 \times 1.5 \times 10^3 \times 9.6}{2 \times 5 \times 10^6}$

$= 9.4 \times 10^{-2}$ m

25. Potential energy per unit volume (E) $= \frac{1}{2} \times$ stress $\times$ strain

$E = \frac{\text{Stress}^2}{2Y} = \frac{(F/A)^2}{2Y} = \frac{F^2}{2A2Y}$

$\therefore \frac{E_1}{E_2} = \left(\frac{A_2}{A_1}\right)^2 = \frac{r_2{}^4}{r_1{}^4} = \left(\frac{3}{1}\right)^4$

$E_1 : E_2 = 81:1$

26. Energy per unit volume (E) $= \frac{1}{2} \times Y \times (\text{Strain})^2$

$= \frac{1}{2} \times 1.2 \times 10^{11} \times \left(\frac{1 \times 10^{-3}}{2}\right)^2 = 15 \times 10^3\ J/m^3$

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